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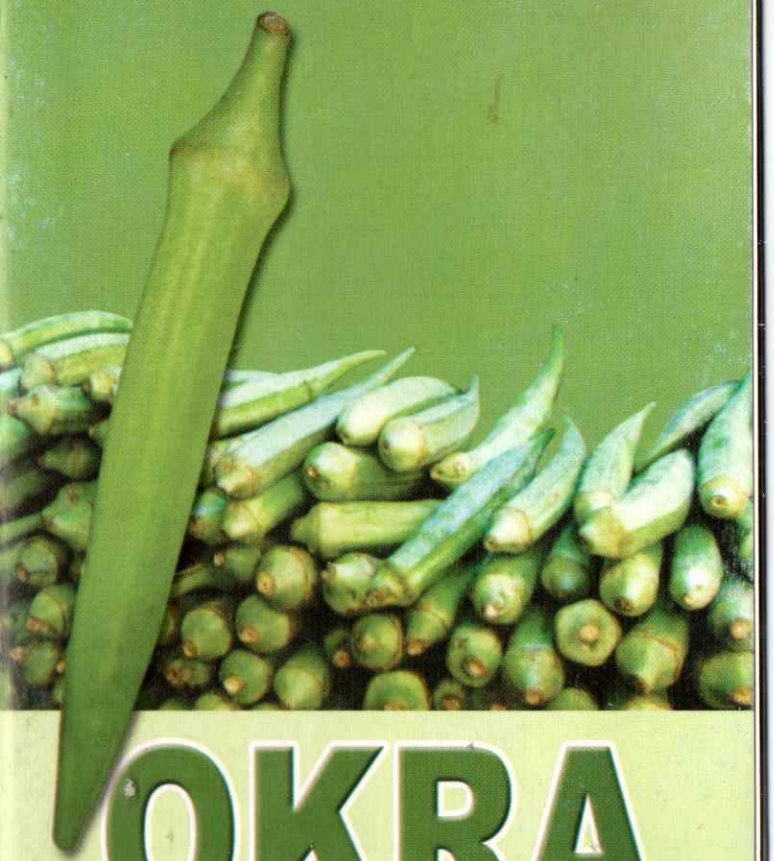

OKRA
Production Guide

Okra (*Hibiscus esculentus L.*) is a tall growing, warm season and annual vegetable crop. It is a popular and a profitable vegetable crop in the country. The young and tender fruits can be prepared as salad, boiled, broiled or fried and can be mixed in any meat and fish dishes. It is also an important vegetable mix of the famous Ilocano dish, *pinakbet*. Okra is rich in vitamin A, protein, calcium, fats, potassium, phosphorus, iron and carbohydrates.

Aside from its nutritional value, okra is used in traditional medicine for the treatment of stomach ulcer, inflammation of the lungs, colitis and sore throat.

Nearly half of the fruit is soluble fiber in the form of gums and pectins. Soluble fiber helps lower serum cholesterol, reducing the risk of heart disease. The other half is insoluble fiber which help keep the intestinal tract healthy decreasing the risk of some forms of cancer, especially colorectal cancer. Nearly 10% of the recommended levels of vitamin B6 and folic acid are also present in a half cup of cooked okra.

Generally, okra is planted for home consumption. But planting this crop extensively can give a year-round income for a farm family.

SOIL AND CLIMATIC REQUIREMENTS

Okra can tolerate a wide range of soil types but for better yield, plant in silty to sandy loam soils that are well-drained and with adequate organic matter.

Okra thrives best in a long, warm growing season.

CULTURAL MANAGEMENT PRACTICES

Selection of Varieties

Chose varieties that are adapted to your local conditions, resistant to pests and diseases and preferred in the local market for better profit.

To guide you in choosing your variety, refer to the *Guide in Selecting Varieties of Lowland Vegetables* included in this kit.

Land Preparation

Prepare the field as early as possible to give enough time for the weeds and stubbles of previous crops to decompose. Plow and harrow 2 to 3 times alternately at one week interval. Plow at a depth of 15 to 20 cm. Harrow twice to break the clods and level the field. A well-pulverized soil promotes good soil aeration and enhances root formation.

A day before planting, prepare furrows at a distance of 75 cm between rows.

Fertilizer Application

It is necessary to apply fertilizers based on soil analysis to ensure the right kind and amount of fertilizer are applied. However, in the absence of soil analysis, follow the following recommendations:

- Aplly 6 to 10 bags of complete fertilizer (14-14-14) and 10 bags of organic fertilizer per hectare. Apply the fertilizer within the furrows and cover with fine soil before planting.

- Sidedress urea (46-0-0) 30 days after planting at the rate of 10 grams or one tbsp per hill.

Planting

Okra is commonly planted directly in the field. It requires 7 to 10 kg of okra seeds to plant a hectare farm. To attain uniform germination and good crop stand, soak the seeds overnight before planting.

Plant okra seeds in slightly moist soil at a rate of 2 to 3 seeds per hill at a distance of 30 cm and 2 to 5 cm deep. Replant missing hills 3 days after emergence.

Irrigation

Water is a limiting factor in crop production. Irrigate or water the plants regularly whenever the soil is dry or the plants show sign of wilting. Use furrow irrigation if available every 7 to 14 days depending on the season and soil type. Thin the plants to two (2) seedlings per hill 15 days after planting. Remove stunted and sickly seedlings leaving only the healthy ones.

Cultivation and Weeding

Off-bar the plants at 15 days after emergence to minimize the growth of weeds. Do not cultivate too close to the base of the plants to avoid disturbing the roots.

Hill-up one month after emergence or 15 days after off-baring to cover the base of the plants. Hand-pull remaining weeds between plants.

Crop Protection

Okra is usually resistant to insect pests and diseases. However, it is necessary to observe the field to make sure that the plants are free from pests.

The most important pest of okra are cotton stainer and stink bug. The most serious fungal diseases are cercospora blight, powdery mildew, fruit rot and root knot nematode.

In case of insect pests infestation follow the recommendations in controlling specific pests given in the *Insect Pest and Disease Management Guide* included in this kit. If diseases are observed, uproot and bury infected plants and maintain the cleanliness of the field.

Maintain field sanitation to prevent pest and diseases occurence.

Ratooning (Optional)

Ratooning is cutting of stem of old plants to induce branching and emergence of new shoots.

Okra can be ratooned to minimize cost of production. After harvesting, cut the stems leaving about one foot from the ground. Shallow cultivate in between rows using plow. Sidedress the ratoon with 10 grams urea per hill to induce shoot emergence. Hill up to cover the fertilizers. Irrigate through the furrows if moisture is not sufficient.

Harvesting and Postharvest Activity

Okra starts to flower 40 to 75 days after planting. Young and tender fruits can be harvested 4 to 6 days from flowering. Immature fruits are more acceptable for table consumption. Harvest only the fruits that measure 3 to 4 inches at 2 to 3 days interval. Harvest in the morning or late in the afternoon to maintain the freshness of the fruits.

A hectare field can produce about 18 to 25 tons of okra fruits.

High quality produce demands better price in the market. Sort and discard malformed and diseased fruits.

Pack okra fruits in woven basket, polyethylene bags, box or wooden crates lined with banana leaves or old newspaper to prevent bruising. Do not overload the containers to prevent compaction of the produce.

COST AND RETURN ANALYSIS FOR OKRA
Per Hectares Basis for CY 2008

PARTICULARS	UNIT	QNTY	UNIT COST (P)	TOTAL COST (0.00)
A. Labor Inputs/Hectare				
1. Land Preparation				
a. 1st Plowing		8	300.00	2,400.00
b. 1st Harrowing		4	300.00	1,200.00
c. 2nd Plowing		5	300.00	1,500.00
d. 2nd Harrowing		3	300.00	900.00
f. Furrowing		2	300.00	600.00
2. Planting	MD	15	150.00	2,250.00
3. Basal Fertilization	MD	5	150.00	750.00
4. Thinning	MD	10	150.00	1,500.00
5. Care of the Plants				
a. Irrigation (2x a wk for 1 month)	MD	9 x 2	150.00	2,700.00
b. Weeding and cultivation (3x)	MD	3 x 5	150.00	2,250.00
c. Side dressing of ferilizer		8	100.00	1,200.00
d. Hilling up	MD	5	300.00	1,500.00
e. Control of pest and diseases	MD	3 x 3	150.00	1,350.00
6. Harvesting (8 primings)	MD	50	150.00	7,500.00
7. Packing	MD	15	150.00	2,250.00
8. Hauling	MD	4	150.00	600.00
Sub-Total				**30,450.00**
B. Materials Inputs				
1. Seeds (OPV)	kg	7	350.00	2,450.00
2. Fertilizer				
Complete (14-14-14)	bag	2	1,800.00	3,800.00
Organic Fertilizer	bag	20	200.00	4,000.00
3. Insecticides	liter	4	850.00	3,400.00
4 Polyethylene	sack	100	60.00	6,000.00
5. Fuel	liter	40	50.00	2,000.00
6. Miscellaneous				1,500.00
Sub-Total				**23,150.00**
Sub-Total (A & B)				**53,600.00**
C. Contingencies (15% of the total labor & material inputs)				8,040.00
GRAND TOTAL				**61,640.00**
Estimated Yield/ha (kg)				**20,000**
Estimated Gross Income (P15.00/kg)				**300,000.00**
Estimated Net Income				**238,360.00**
Return on Investment (ROI)				**386%**
Break Even Price (per kg)				**11.94**

MD man days

REFERENCES

OKRA PRODUCTION GUIDE. Agricultural Communication Section, Department of Agriculture Regional Field Unit No. 02, Tuguegarao City.

COST OF PRODUCTION PER HECTARE OF DIFFERENT VEGETABLES: ISABELA CONDITION. 2004. A Handout. HVCC Section, Department of Agriculture Regional Field Unit 02, Tuguegarao City.